Observation is Theory-Laden Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 25-35 min

Category	Natural Sciences	**Time**	25-35 min
Difficulty	Medium	**ID**	observation-is-theory-laden

Big idea: Seeing scientifically requires concepts.

One-Minute Teacher Brief

Students see that observation depends on concepts, training and context.

Student Prompt

Inspect Observation is Theory-Laden Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Observation is Theory-Laden Stimulus A	Student-facing stimulus 1: The same candle flame described by a poet, chemist, safety inspector, and historian. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Observation is Theory-Laden Stimulus B	Student-facing stimulus 2: A simple phenomenon viewed with different guiding questions. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Observation is Theory-Laden Reveal / Twist	Student-facing stimulus 3: A notice-and-ignore table showing how theory directs attention. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Seeing scientifically requires concepts.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Observation is Theory-Laden Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Observation is Theory-Laden Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Can observation be theory-free?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how seeing scientifically requires concepts.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

• Work silently for the first response so you can notice your own starting point.

• Record one knowledge claim, the evidence behind it, and your confidence level.

• After the reveal or comparison, update your claim in a different colour or column.

• Use the debrief to answer: Can observation be theory-free?

Setup Checklist

• Prepare a starter stimulus using: The same candle flame described by a poet, chemist, safety inspector, and historian.

• Print or project the student response grid before the activity begins.

• Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

• The same candle flame described by a poet, chemist, safety inspector, and historian.

• A simple phenomenon viewed with different guiding questions.

• A notice-and-ignore table showing how theory directs attention.

Resources To Use

• Resource pack: resources/observation-is-theory-laden/index.html

• Card deck CSV: resources/observation-is-theory-laden/cards.csv

• Visual prompt: resources/observation-is-theory-laden/visual-prompt.svg

• Printable student worksheet with observation, inference, evidence, and confidence columns.

• Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.

• Classroom run kit with prompt cards, example stimuli, and debrief moves.

• PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Show an image without context and ask students what they observe.

2. Now label it as a microscope image, satellite image, medical scan, particle trace or climate graph.

3. Ask what new things they notice.

4. Discuss whether the image changed, or the knower changed.

Debrief Moves

• Knowledge question: Can observation be theory-free?

• Does training help scientists see more, or see differently?

• What role do models play in scientific knowledge?

Teacher Quick Script

• Write your first judgement before we discuss it; the first answer is useful evidence.

• Separate what you noticed from what you inferred.

• What would make this knowledge claim more reliable?

Exit Ticket

• One thing I first treated as evidence was...

• My confidence changed because...

• A better knowledge question I can now ask is...